'Murder hornets' have arrived in the U.S.—here's what you should know

by Douglas Main, nationalgeographic.com May 4, 2020

Update on May 29, 2020: Washington State officials have confirmed the discovery of a third Asian giant hornet, almost certainly a queen, found dead on a road near Custer, Washington. This likely means a colony was present in the state in fall 2019, and that it produced many queens. These large females lie dormant over the winter, and a small percentage are likely to create successful nests. The finding adds urgency to ongoing efforts to trap and track the hornets, said Sven Spichiger, an entomologist with the Washington State Department of Agriculture, in a news conference.

Two unusual hornets—striking, with orange and black markings and long stingers—were spotted near Blaine, Washington, in late 2019. Subsequent investigation revealed they were Asian giant hornets, the world's largest wasps, growing nearly two inches in length.

Scientists are concerned that these hornets could spread throughout Washington State and beyond, presenting a danger to U.S. bees —which are already in decline—and humans.

Nobody knows how the insects arrived in the United States. But the discoveries set off alarms and the insects began trending on social media as "murder hornets." These predators, native to East Asia and Japan, are infamous for decimating honeybee colonies.

With the toxic venom that their large stingers deliver, the insects already are known for killing people in their native habitats: In Japan,

an average of 30 to 50 people each year die from the hornets' stings. In 2013, when populations of the hornets were unusually high, they killed 42 people in a single Chinese province. Most serious incidents occur when people come near or disturb the insects' hives.

These insects "are pretty formidable," says Chris Looney, an entomologist at the Washington State Department of Agriculture. "I am very worried."

At the same time, Looney offers some caution, starting with the nickname "murder hornet." He hadn't heard it before recent press coverage of the insects—and he doesn't love it.

"I worry people are already scared enough of insects for dubious reasons," Looney says. But he grants that there's an upside to the ominous label: "It does seem to have gotten people's attention. I just hope the sensational 'murder hornet' coverage helps us understand our ecosystems a little better."

Voracious feeders

As of now, researchers can't confirm how the hornets arrived. Looney says it's most likely they got accidentally trapped in shipping containers from one of the countries where they're native.

One complete hive of the insects was found and destroyed in late 2019 in nearby Nanaimo, Canada. But genetic tests suggest those giant hornets were introduced separately.

Asian giant hornets from Japan and East Asia already have

established themselves as invasive species in other nations, such as South Korea. These social wasps form colonies, comprising one queen and many workers, which can fly half a dozen miles or more from the hive to find food. They eat many kinds of insects, but they seem to especially enjoy feasting on bees.

When they encounter honeybees, their attack starts with a “slaughter phase” in which they serially bite the heads off bees with their large mandibles, Looney explains. Within 90 minutes, a small group of Asian hornets can destroy an entire colony’s workers this way, he says.

Then, the hornets shift to feeding. They occupy the honeybee nest for up to a week or longer, feeding on the pupae and larvae. They then feed it to their own young.

European honeybees (*Apis mellifera*), the most widespread commercial pollinators, have no known defense against Asian giant hornets. Though the bees have been observed stinging the would-be invaders, it appears to have no effect on these wasps.

By contrast, Japanese honeybees (*Apis cerana japonica*), which have co-evolved with giant hornets, have found a way to protect themselves. By swarming around the invaders and beating their wings until this “bee ball” reaches temperatures over 115 degrees Fahrenheit, the bees cook the wasps to death and suffocate them with carbon dioxide.

Asian giant hornets could also have deadly impacts on pollinators like native bee species, many of which are already suffering from competition with other exotics, Looney says.

Jun-ichi Takahashi, a researcher and wasp expert at Kyoto Sangyo University in Japan, agrees the wasp's damage to the environment could be significant anywhere they spread—and supports efforts to contain it.

Interviewed by email, Takahashi says the label murder hornet is appropriate because the species is as dangerous as killer bees, a nickname for the African honeybee. That invasive bee species arrived in Texas in 1990 and has colonized parts of the southern U.S. and caused human deaths.

It's Takahashi's opinion that "Americans do not fully understand the aggressiveness and toxicity of this hornet."

Stopping the spread

If the hornets aren't headed off in the next couple years, it will probably be too late to halt their spread across the United States, Looney says.

As of now, there have been only two sightings of worker bees of the species near Blaine, which suggests that there is likely a colony nearby. In winter, nests go dormant, and if any queens have mated, they would disperse to form new hives. Currently, Looney and other researchers are setting out lures to try to capture emerging queens.

In the summer, the researchers will set out hundreds of traps to continue looking for queens and workers, which would emerge in the summer if any new colonies are established. They could then try to attach radio-transmitting collars so they could track the wasps back to

their nests and destroy them, Looney says.

Because the hornets make hives underground that generate heat, Looney and colleagues are also experimenting with heat-sensitive imaging technologies to help them find the hives.

Still, “It's going to be tough” to stop the hornet, Looney says. “But yeah, we have a shot at it.”

Know your hornets

Asian giant hornets have stingers a quarter-inch long, which can pierce protective clothing that beekeepers normally wear. Research shows that even in people who aren’t allergic, 50 or fewer stings can cause death due to kidney damage.

Since the hornets made the U.S. news, Looney has been inundated by emails from concerned Americans who think they’ve seen the insect. In response, he stresses that only two sightings of the Asian giant hornet have been confirmed in the U.S.—and that people who think they’ve seen it in eastern North America are likely confusing it with the European hornet (*Vespa crabro*), which looks similar but is less aggressive and dangerous.

“If people east of the Mississippi see something like that, they shouldn't assume it's an Asian giant hornet,” he says. “It's almost certainly not.”